308 Approach to the Patient with Renal Disease or Urinary Tract Disease

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The upper urinary tract consists of the kidneys, their vasculature, and the renal parenchyma and its collecting system. The lower urinary tract is composed of the peristaltic ureters from each kidney, the bladder which receives and stores urine from each ureter, and the urethra which, upon bladder contraction, excretes the final urine. The excretory system as a whole begins with a plasma ultrafiltrate from renal cortical glomeruli, modified by the renal tubules of the nephron, and ends with the excretion of urinary water and solutes.

Disease can occur at any level of these functional structures with either no symptoms or signs (such as an incidental renal mass picked up on ultrasound); with nonspecific findings (such as fatigue); with signs highly specific for a syndrome of dysfunction referable to a structure, but not diagnostic of a particular disease (such as proteinuria); or with a finding highly characteristic of a specific diagnosis (such as polycystic kidney disease). Disorders of the kidney presenting without symptoms (such as invisible microhematuria or a stone) are often discovered by laboratory testing or imaging.

It is important to perform renal testing in patients with systemic disease (for example, routine screening for albuminuria in diabetes mellitus [DM]). Likewise, one should look at renal function when a patient's history includes known risk for renal or urinary tract complications (for example lithium for psychiatric treatment, occupational use of lead, or, in the case of bladder cancer, exposure to aniline dyes). This chapter will address an approach to patients with and without known history of kidney disease, as well as those who have one of a wide variety of systemic illnesses that involve the kidney or the lower urinary tract.

To begin with the evaluation of renal disorders, note that it is time-consuming and wasteful to do random testing prematurely, which would generate a very long list of potential diagnoses. A better approach is to look at the evidence from the history, physical examination, and basic laboratory studies to arrive at a logical basis for targeted tests that would identify a particular disease process (Table 308-1). The findings are influenced by the regions of the kidney that are involved and by factors such as family history, toxic exposures, the patient's birth weight and age, and time course of the findings. As a rule, kidney diseases that begin primarily with glomerular dysfunction have albuminuria as the hallmark, whereas those kidney diseases that begin in the tubular structures may present primarily with electrolyte disorders or disorders of dilution and concentration of the urine. In both glomerular and tubular disorders, upon progressing to chronic disease, the distinction is more difficult because glomerular diseases eventually affect the tubular interstitium and tubular diseases progress to glomerular dysfunction and scarring.

One example of the progression from a tubular disorder to chronic kidney disease with late glomerular damage is the lysosomal storage disease cystinosis, one of the childhood Fanconi's syndromes. Within the first year of life, the affected child may become easily dehydrated from salt-wasting; feed and grow poorly; develop polyuria, hypotension, and muscle weakness; and show the features of proximal tubular dysfunction. Electrolyte losses result in hypokalemic renal tubular acidosis (RTA), renal glycosuria, phosphaturia causing growth delay from renal rickets, and acidemia. The accumulation of lysosomal cystine leads to destruction of the proximal tubule and adjacent interstitium, while glomerular filtration remains near normal and urine contains little albumin.

Even with treatment, progressive scarring of the capillaries within the interstitial space eventually, over a decade or more, leads to albuminuria and progressive decrease in GFR. Electrolytes like K^+ and $HPO_4^=$ are retained at this stage and metabolic acidosis results from failure of the kidney to produce NH_4^+ rather than losses of HCO_3^-. Hypertension and edema from salt and water retention replace the low blood pressure associated with the loss of fluid at the earlier stage of tubular dysfunction. Anemia results from loss of erythropoietin production in chronic kidney disease. The symptoms of weakness and fatigue are nonspecific in that they might correlate with anemia, hypo- or hyperkalemia, hypophosphatemia, acidosis, l-carnitine deficiency from proximal tubule reabsorptive failure in Fanconi's syndrome, or azotemia. Hepatomegaly on a newborn examination, typical of but not specific to cystinosis, might raise suspicion of other inborn errors of metabolism such as glycogen storage diseases. However, the finding of photosensitivity related to highly refractile cystine crystals deposited in the cornea is very specific for cystinosis.

In the following discussion, the focus is on symptoms and signs that constitute the major syndromes seen in the patient with kidney or lower urinary tract disease. These syndromes are the foundation for the diagnosis of particular diseases.

TABLE 308-1 Acute Kidney Injury

FEATURES	PRERENAL	RENAL	POSTRENAL
CONTEXT	Heart failure, hepatic failure, burns, shock, post-op, dehydration, renovascular disease, vascular drugs	ATN, bilateral cortical necrosis, AIN, crush injuries, myoglobulinuria, hemoglobinuria, ischemia, sepsis, recent contrast drugs	Bladder outlet obstruction, obstruction of solitary kidney, abdominal mass, bilateral stones, drugs
URINE OUTPUT	Oliguria (usually <500 mL/day) Anuria	Oliguria Normal Polyuria	Anuria Polyuria Both
CLINICAL HISTORY AND EXAM	Trauma, shock, hypotension, burns, GI, sweat, or renal losses	Rhadomyolysis, hemolysis, thrombotic microangiopathy, AIN, atheroemboli	Extrinsic ureteral obstruction, retroperitoneal disease, bladder outlet or urethral obstruction by prostate or cervical cancer
URINALYSIS	Concentrated acid urine, hyaline casts, crystalluria	"Muddy brown" granular casts, hematuria, dysmorphic RBCs, RBC casts in GN, WBC casts in AIN, eosinophils in atheroemboli	Intrarenal obstruction by uric acid or calcium phosphate in tumor lysis syndrome, blood clots in lower urinary tract bleeding, CaOx after ethylene glycol ingestion
URINE CHEMISTRY	Low FENa ≤1%, U/P Cr ≥40, UNa ≤10, U/P Osm ≥1	FENa 1–3%, U/P Cr ≤40, UOsm ~Isomotic	Early looks like prerenal, late looks like renal
LABORATORY STUDIES	High BUN/Cr ratio, usually hypercatabolic with increased uric acid; may have hyper- or hyponatremia	May have eosinophilia in allergic AIN, high phosphate, low Ca, high PTH, metabolic acidosis	Hydronephrosis on ultrasound, extrinsic or instrinsic diseases on CT scan, tumors on MRI

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NEPHRITIC SYNDROMES

Nephritis literally means an inflammatory condition of the kidney. Nephritic inflammation may occur in association with infection, allergic response to medications, systemic autoimmune disorders, or toxic exposures. The kidneys may be one of many organs, or the only organ, involved in the inflammatory disorder. Inflammation with enlargement of the kidneys is often associated with point tenderness over the flank and sometimes exquisite costovertebral angle tenderness, requiring gentle palpation for diagnosis.

The nephritic syndromes may be further subdivided depending on the inflamed structures within the kidney, including vascular, glomerular, and tubulointerstitial; and also, on the time course of progression of the inflammatory process, which may be acute, subacute, or chronic.

GLOMERULONEPHRITIS

Glomerulonephritis (GN) is associated with hypertension, volume expansion, and an abnormal urinalysis. In most cases, the volume expansion manifests as edema and hypertension; in the child, ascites may develop; and in the elderly, restlessness and anxiety may be the first signs of incipient acute pulmonary edema. There may be orthopnea at night or dyspnea on exertion, with or without significant peripheral edema. Acute GN is usually associated with low urine output (oliguria or at times complete anuria), low urine sodium content, and concentrated urine, resulting in the retention of salt and water. It is an absolute necessity to observe the urine, including a spun urinary sediment, to diagnose active GN.

The disease presentation may be acute or subacute (postinfectious GN and rapidly progressive GN), occurring over days or weeks; or it may be chronic, occurring over many months to years. The pathologic correlate of these presentations—the relative amount of acute inflammatory, proliferative, or necrotizing lesions in renal tissue versus chronic sclerotic and atrophic findings on renal biopsy—reflects the acuity versus chronicity of disease. Therefore, in these progressive destructive conditions, early diagnosis is critical and treatment for acute reversible disease should be instituted as quickly as possible.

The evaluation of the findings on history, physical examination, and urinalysis in narrowing diagnostic possibilities for a specific etiology of the glomerular syndromes is demonstrated in **Fig. 308-1.** The nephritic urinalysis shows hematuria, proteinuria, cells or clumps of cells, and cellular casts in the spun urinary sediment. The cells in the urine are usually a mixture of red blood cells and inflammatory cells, including polymorphonuclear (PMN) leukocytes. Microscopic hematuria is invisible to the naked eye, as opposed to macroscopic hematuria, identifiable as cola or "tea-colored" urine, caused by hemoglobin entering an acid urine. The red cells enter the renal tubules through breaks in the basement membrane of glomeruli. The process of moving across the membrane causes erythrocytes to become misshapen or dysmorphic. It is unusual in these nephritic syndromes to see blood clots or gross hematuria, a term used for overtly blood-red urine that characterizes other syndromes or lower urinary tract bleeding. However, in episodes of IgA nephropathy, glomerular hematuria is often "gross" as RBCs traverse rents in the glomerular basement membrane (GBM).

Postinfectious Glomerulonephritis Postinfectious glomerulonephritis (PIGN) is the classic example of acute GN, a complement-mediated immune complex response, to a bacterial antigen occurring 10 days to 3 weeks *after* a specific nephritogenic strain of group A streptococcal pharyngitis, or *after* skin infection such as impetigo. Because the sore throat or skin infection may have already resolved, antibiotic treatment may not be necessary at this stage of renal complication, but it is important for the clinician to inquire about these possible prior events. Subacute bacterial endocarditis, if present by history, may also result in a circulating immune complex disorder, even while the patient remains on antibiotic treatment. These glomerulonephritic and other immune complex conditions are recognized by low-C3 complement levels in the blood. However, the postinfectious cases should be distinguished from cases of infection-associated glomerulonephritis that occur *during* an ongoing infection, such as a staphylococcal (often MRSA) abscess or empyema, which are treated with antibiotics and purulent drainage and are characteristically associated with IgA and complement deposition in the glomeruli.

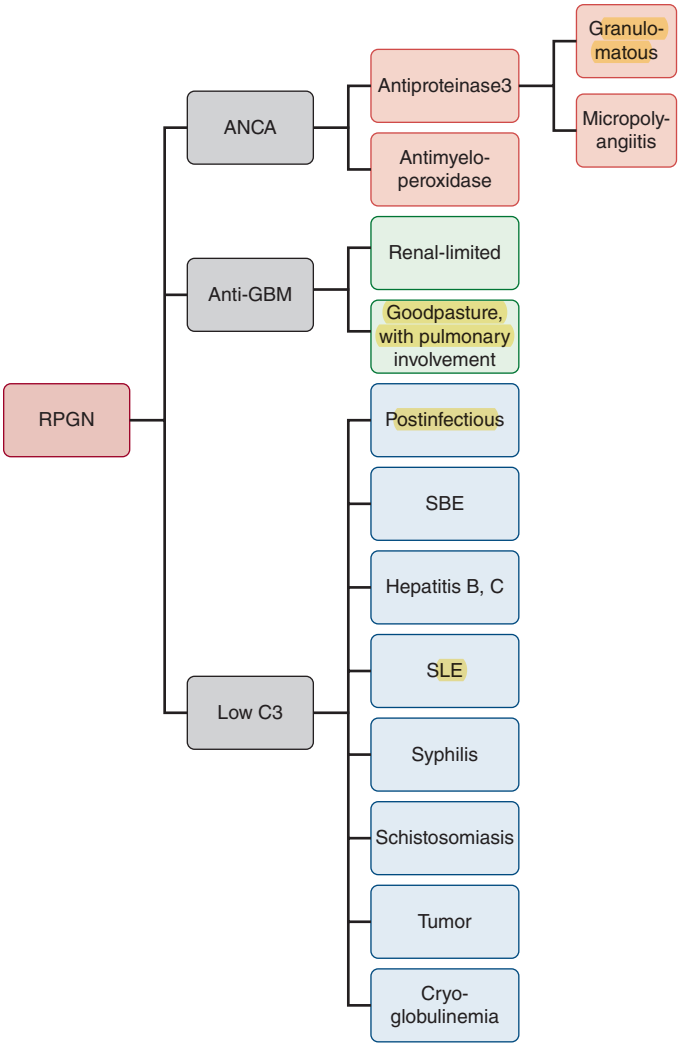


FIGURE 308-1 Rapidly progressive glomerulonephritis. The syndrome of RPGN is diagnosed clinically. The three major categories are the antineutrophilic cytoplasmic antibodies (ANCA-positive vasculitities); antiglomerular basement membrane antibody-positive (anti-GBM); and immune complex disorders with low C3 complement. C3 is synthesized in the liver, bound to circulating infectious or neoplastic antigens with their antibody complexes, and deposited in the glomerular subendothelium is associated with non-renal manifestations. Syphilis is usually accompanied by vasculitis, and cryoglobulins may be in setting of hepatitis C or myeloma and often lowers C4 as well. SBE, subacute bacterial endocarditis; SLE, systemic lupus erythematosus.

Postinfectious GN is to be distinguished from synpharyngitic hematuria, another glomerular syndrome that frequently produces gross hematuria, a heavier excretion of red blood, which may appear to the patient as a frightening hemorrhage. The patient may present with this concern so it is important for the clinician to recognize that as little as 10–20 mL of blood will turn a liter of urine red. Synpharyngitic hematuria, usually with a viral pharyngitis, is most often related to IgA nephropathy rather than postinfectious GN. Another feature that distinguishes IgA nephropathy from postinfectious GN is that the former usually demonstrates normal circulating C3 complement as opposed to low complement in PIGN. IgA nephropathy can cause chronic microhematuria or bouts of gross hematuria or can occur in association with other immunologic conditions including celiac disease, rheumatoid arthritis, reactive arthritis, and ankylosing spondylitis. IgA nephropathy and staphylococcal-associated GN are more common in Asian populations.

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS (RPGN) Rapidly progressive glomerulonephritis (RPGN) is a syndrome arising from a variety of causes (Fig. 308-1). Pathologically it is associated with a proliferation of glomerular parietal epithelial cells and inflammatory cells called cellular *crescents* surrounding the capillary that, over time, becomes fibrotic and atrophic with global loss of the glomerular tuft.

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When >50% of glomeruli are affected (diffuse), this highly destructive process usually leads to glomerular sclerosis. Recognition of RPGN and choice of appropriate therapy are extremely important because, if left unchecked, the syndrome can lead to complete and irreversible loss of kidney function, as well as fatal pulmonary hemorrhage when associated with lung vasculitis.

In the case of RPGN, one sees the typical signs of the nephritic syndrome, though other clues to diagnosis may be present in the pulmonary-renal syndromes, where the patient may present with hemoptysis, interstitial lung disease, epistaxis, or upper airway symptoms of sinusitis or nasal congestion. Blood testing helps narrow the differential diagnosis. The antineutrophil cytoplasmic antibody (ANCA) test, particularly those made up of antimyeloperoxidase (pANCA) or antiproteinase-3 antibodies (cANCA), is diagnostic for the pauci-immune disorders of systemic vasculitis. Granulomatous vasculitis (formerly known as Wegener's granulomatosis) and microscopic polyangiitis are conditions particularly associated with pulmonary and upper respiratory disease. By contrast, a positive test for antiglomerular basement membrane (anti-GBM) antibodies would be consistent with either renal-limited anti-GBM disease or, when associated with pulmonary hemorrhage, with Goodpasture disease. The latter is more often seen in young males who smoke or have a history of inhaling hydrocarbon solvents.

The immune complex diseases that manifest as RPGN have low C3 levels secondary to high clearance of circulating complement with immune complex deposits in the glomerular subendothelial space. Certain of these disorders, such as systemic lupus erythematosus (SLE) and cryoglobulinemia, can present with pulmonary hemorrhage. Patients with cryoglobulinemia often have low C4 complement and a high rheumatoid factor, and the syndrome may be caused by paraproteinemias or hepatitis C. Another cause of crescentic RPGN but with normal C3 complement is a form of IgA vasculitis (Henoch-Schonlein purpura), a syndrome of skin vasculitis characterized by palpable purpura, gastrointestinal bleeding, and arthralgias, and in the vasculitic form, pulmonary hemorrhage. The C3 complement (an acute phase reactant) is detected in the kidney biopsy but not associated with low serum levels. The diagnostic tests and distinct diseases associated with these immune complex disorders are shown in Fig. 308-1.

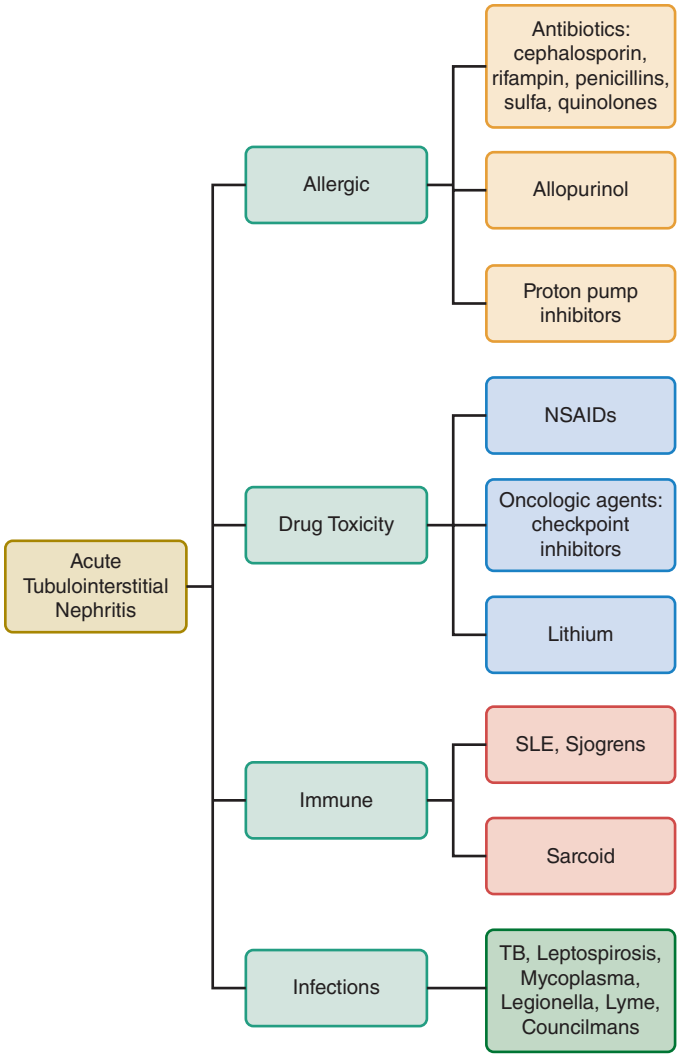


FIGURE 308-2 Acute tubulointerstitial nephritis. Diseases resulting from injury to the tubular and interstitial components of the renal cortex and medulla.

Tubulointerstitial Nephritis Tubulointerstitial nephritis (TIN) comprises inflammatory disorders of the renal tubules and interstitium, which may be caused by infection, autoimmune disease, allergic immunologic responses to certain drugs (Fig. 308-2) and have a time course ranging from days to weeks and months.

ACUTE ALLERGIC AND IMMUNE INTERSTITIAL NEPHRITIS Acute allergic or immune interstitial nephritis (AIN) usually occurs 1 day to 2 weeks following exposure to an offending drug and may be associated with a rapid and potentially reversible loss of kidney function, which may occur in the setting of a change in dose or the restarting of a previously used medication. Associated glomerular proteinuria sometimes occurs with the use of nonsteroidal anti-inflammatory drugs (NSAIDs) or ampicillin. Clinically, there may be fever, rash, and eosinophilia; the last is typical for certain penicillins, fluoroquinolones, and some biologic cancer drugs that act as checkpoint inhibitors (CPIs) but is atypical for NSAIDs. Patients who recover from CPI-induced acute kidney injury (AKI) may be restarted on the inhibitor.

The urinalysis usually shows pyuria and at times eosinophiluria, but the most characteristic cell types are activated T lymphocytes and plasma cells, along with some white blood cell casts. A cyto-centrifuged specimen containing the sediment within a small tube mounted perpendicular to the slide, then stained with Giemsa, best demonstrates these cell types.

The patient may experience symptoms of polyuria and tender kidneys, and signs of tubular dysfunction including nephrogenic diabetes insipidus, hypo- or hyperkalemia and hyperchloremic metabolic acidosis. Common drugs that cause AIN include the proton pump inhibitors (PPIs) and sulfa drugs, especially sulfamethoxazole, but also extending to sulfa-containing diuretics such as acetazolamide, thiazides, and furosemide, and less frequently, bumetanide. PPI's may increase risk for AIN in patients on check point inhibitors. Frequently, rifampin and a few other drugs cause a noncaseating granulomatous interstitial nephritis; in caseating granulomas resulting from hematogenous spread of mycobacterium tuberculosis, the first granulomas appear connected to the glomeruli in the cortex where oxygen tension is at its highest. Granulomatous TIN may occur in sarcoidosis.

Systemic infections including bacterial, viral, fungal, and parasitic may induce tubulointerstitial nephritis, first described pathologically in the preantibiotic era as Councilman's nephritis in the course of scarlet fever.

Autoimmune interstitial nephritis is seen in diseases such as lupus nephritis, ANCA-positive vasculitis, and other rheumatic disorders including Sjogren's syndrome, which may present with dry eyes as part of the sicca syndrome. In a patient with photophobia and red painful eye, the clinician should look at renal function and the urinalysis for evidence of tubulointerstitial nephritis uveitis (TINU) syndrome.

Noninflammatory Interstitial Diseases Noninflammatory interstitial diseases are often caused by toxic exposures that damage the tubular interstitial structures. For example, a cancer patient on ifosfamide may develop Fanconi's syndrome, indicating proximal tubule damage. Heavy-metal exposures such as cadmium, lead, and mercury from old dental fillings may lead to proximal tubule injury and, again, Fanconi's syndrome. A patient with hypomagnesemia and acute kidney injury (AKI) may have had past exposure to platinum-containing chemotherapeutic agents and a patient with nephrogenic diabetes insipidus, unresponsive to antidiuretic hormone, may have

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had treatment with lithium, analgesics, or chemotherapy. Hypokalemia and interstitial disease are consistent with a past history of exposure to aminoglycosides or amphotericin-B. The patient with the former may complain of hearing loss and the patient with the latter may present with hypokalemia and RTA.

Chronic interstitial diseases should be suspected in cases of paraproteinemia (light-chain nephropathy or amyloidosis) and in patients who ingest herbal remedies containing aristolochic acid, as occurs commonly in China and has been determined to be the cause of Balkan nephropathy. Perhaps the most common cause of chronic interstitial nephritis is prolonged analgesic use to treat chronic pain (not only NSAIDs but acetaminophen or combinations of phenacetin, aspirin, and caffeine) as part of the analgesic syndrome. The clinician should ask about a prior history of pain and also gastrointestinal symptoms that may precede the kidney disease; the analgesic drugs often go unreported by the patient. It is important to recognize this syndrome because, if the drugs are not discontinued, a later development may be urothelial cancers of the distal ureter and urinary bladder. Lithium after prolonged use can also cause chronic interstitial nephritis along with nephrogenic diabetes insipidus and slowly progressive kidney failure.

Patients who have hypercalcemia or hyperoxaluria may develop nephrocalcinosis, a form of interstitial nephritis characterized by calcifications within the renal parenchyma, often at the cortical medullary boundary. When nephrocalcinosis and nephrolithiasis are concurrent, the most common etiologies include hypercalcemic disorders, particularly primary hyperparathyroidism, and congenital medullary sponge disease (tubular ectasia), as well as hereditary distal RTA. In patients with DM, chronic analgesic use, or sickle-cell diseases, the phenomenon of papillary necrosis is associated with chronic interstitial damage characterized by nephrogenic diabetes insipidus. For example, phenacetin toxicity is a result of the drug's concentrating in the medulla due to the normal urinary concentrating mechanism; ironically, the first function lost because of medullary toxicity is the ability to concentrate the urine. In the ensuing papillary necrosis, the patient may observe solids in the urine, which are sloughed tissue from the ischemic medulla. When a patient with inflammatory bowel disease or one who has had gastric bypass procedures such as the Roux-en-Y develops kidney injury, with or without calcium oxalate kidney stones, a 24-h urine oxalate measurement must be determined to diagnose intestinal hyperoxaluria, which otherwise may lead to nephrocalcinosis. Chronic bacterial pyelonephritis is not a common cause of chronic renal failure because it rarely affects both kidneys. Acute pyelonephritis in a solitary kidney or transplanted kidney may cause AKI.

PROTEINURIC STATES AND NEPHROTIC SYNDROME

Proteinuric States Some patients notice generalized edema and foamy urine, manifestations of proteinuria. Proteinuria is not synonymous with nephrotic syndrome (NS). In DM the urinary loss of microalbuminuria (MALB) defines glomerular proteinuria, which typically occurs after years of small-vessel disease (explaining the likelihood of coincident small-vessel retinopathy) and predicts future development of progressive renal failure and NS. The rate of renal progression is greater with hypertension, obesity, and poor glucose control. Further risks are nephrectomy, hepatitis C, and, as in focal segmental glomerular sclerosis (FSGS), in patients of African ancestry is likely due to the prevalence of the APOL1 gene mutations. Because treatment of patients with MALB is indicated, the clinician caring for diabetic patients should screen regularly for the presence of small amounts of albumin in the urine, which can be detected as microalbuminuria (30–300 mg/day of albumin excretion) by radioimmunoassay. Subnephrotic albuminuria is characteristic of focal diseases of the kidney with <50% of glomeruli involved, whereas NS is likely diffuse, involving most glomeruli. Lower levels of albuminuria are characteristic of glomerulonephritis, as discussed above.

Microalbuminuria is beneath detection by urinary dipstick protein analysis. Any detectable albuminuria on dipstick is called *overt proteinuria* and when detected at the highest level on the strip is consistent with nephrotic proteinuria. The dipstick measurement picks up only the most acidic proteins like albumin but not proteins with a higher isoelectric point, most notably kappa and lambda light chains found in multiple myeloma. Additionally, the filtration of light chains may cause glomerular damage and albuminuria (kappa light-chain nephropathy or lambda AL amyloid) but does not require an abnormal glomerulus because their charge and molecular weight allow them to freely cross the glomerular barrier. Such filtration is considered overflow proteinuria.

A laboratory measurement of albumin or total protein should be normalized to urine creatinine concentration as a ratio in order to discount the effects of urinary dilution or concentration. It is also important to note that the ratio of total protein concentration to creatinine concentration is not specific for albumin excretion and may indicate the presence of light chains. The detection of light chains requires a urine protein electrophoresis.

Another type of proteinuria known as tubular proteinuria, which is mostly β-2 microglobulin, is secreted by proximal tubular cells and is common in interstitial nephritis.

Nephrotic Syndrome Nephrotic syndrome (NS) has three defining features: edema, hypoalbuminemia (<3.5 g/dL), and proteinuria >3.5 g/day. The syndrome is often associated with lipid abnormalities such as a high LDL, low HDL, and lipiduria. The urine may contain large tubular epithelial cells that are engulfed with the lipid in a recognizable shape under polarized light. These oval fat bodies may also contain cholesterol monohydrate crystals, which appear as Maltese crosses. Under low light, the refractile lipid may be seen as fatty casts.

Clinically, patients present with generalized edema and, in contrast to heart failure with orthopnea, facial, eyelid and periorbital swelling is observed in NS. Penile and scrotal edema can be severe enough to obstruct urethral urine flow in men. In some cases, edema may form large bullae that may rupture, predisposing to ulceration and cellulitis. The skin becomes smooth and may appear to "weep." Patients may have hoarseness caused by vocal cord edema. On occasion, nephrotic range proteinuria is not associated with edema, for example in the case of human immunodeficiency virus-associated nephropathy (HIVAN), which is more common in blacks with HIV. In these cases, rapid loss of renal function may occur due to implosion of the glomerular capillary by proliferation of visceral epithelial cells that crush the glomerular loops, decreasing capillary flow and filtering surface area (collapsing glomerulopathy). This glomerulopathy is a severe, treatment-resistant form of FSGS. Other secondary forms of FSGS, more often not associated with NS, include remnant kidney after partial surgical removal, adaptive injury, chronic hypoxemia, sickle cell diseases, recurrent vasculitis with healing of some glomeruli, obesity, and congenital urogenital anomalies. Very heavy proteinuria is often noted in this syndrome. Secondary forms of FSGS are shown in **Fig. 308-3**. Other secondary forms of FSGS, more often not associated with NS, include remnant kidney after partial surgical removal, adaptive injury, chronic hypoxemia from chronic lung or congenital heart disease, sickle cell diseases, recurrent vasculitis with healing of some glomeruli, obesity, and congenital urogenital anomalies.

NS is a hypercatabolic state with negative nitrogen balance due to proximal tubule absorption and lysosomal catabolism exceeding hepatic albumin synthesis. A characteristic finding of rapid-onset hypoalbuminemia in NS is horizontal linear white lines in the nail bed, known as Muehrcke's lines. When NS in the adult occurs abruptly, with severe elevation of cholesterol, one must consider glomerular epithelial injury (podocytopathy), which may be idiopathic minimal change disease (MCD), a name historically based on the appearance of renal tissue on light microscopy and often preceded by an upper respiratory infection, allergies, or immunization. Secondary causes of MCD are Hodgkin's lymphoma or other lymphoproliferative disorders. MCD can occur at any age and in the adult may present with AKI as well, primarily because of the underlying vascular disease and hypoalbuminemia.

Age is important in the onset of NS, in that young children <6 years of age frequently have MCD. Several gene mutations encoding for proteins of the podocyte, such as nephrin (*NPHS1*) and podocin (*NPHS2*)

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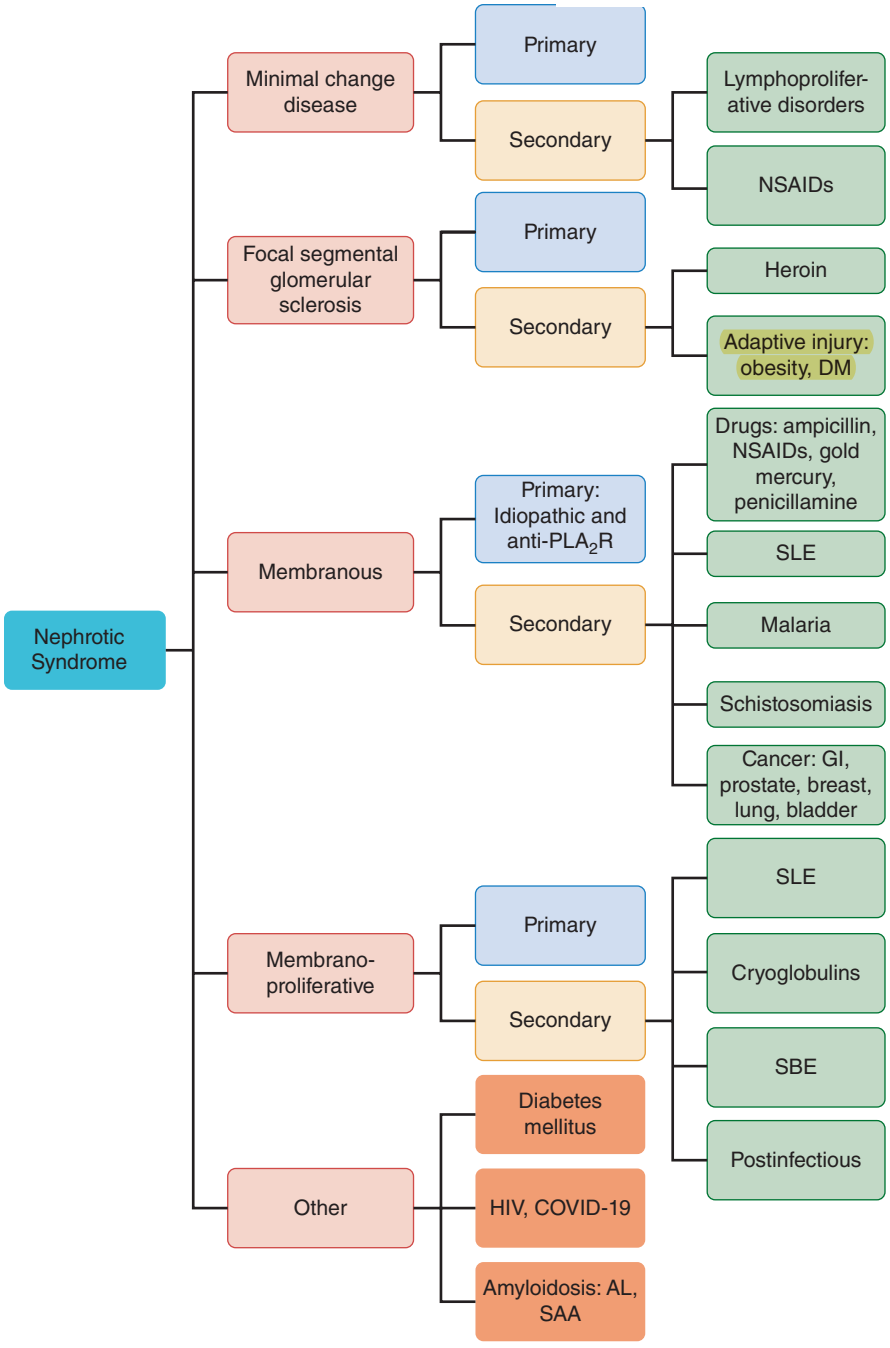


FIGURE 308-3 Nephrotic syndrome. Different pathologic syndromes broken down into the idiopathic and secondary causes. 70% of primary membranous nephropathy (MN) is associated with antibodies to the integral subepithelial basement membrane antigen phospholipase A_2 receptor (PLA_2R). The former term for primary membrano-proliferative GN has been replaced by pathologic syndromes involving complement deposition (C3 glomerulopathy and dense-deposit disease in children). AL amyloid is the form of amyloidosis secondary to lambda light-chain deposition and SAA is the protein (serum amyloid A) associated with chronic inflammatory diseases such as rheumatoid arthritis, familial Mediterranean fever, and tuberculosis.

in the slit diaphragm, have been shown to be responsible for most cases of hereditary nephrotic syndrome (see **Table 308-2**).

In children who have a serum albumin concentration <2 g/dL, tissue ischemia may develop, causing a "nephrotic crisis," in which severe abdominal pain may be mistaken for a surgical abdomen. A 20-year-old presenting with NS may have MCD, FSGS, or membranous nephropathy (MN).

Like MCD, MN exists as primary or secondary etiology. Whereas NS is a cause of a hypercoagulable state, particularly when it is accompanied by severe hypoalbuminemia (<2 g/dL), MN is the NS most often associated with renal vein thrombosis (RVT) for a number of reasons, among them the urinary loss of antithrombin-3 and plasminogen, as well as hyperfibrinogenemia. RVT may present with back pain or pulmonary emboli.

NS with heavy proteinuria or large kidneys on ultrasound in advanced stages of chronic kidney disease (CKD) is often noted in diabetic nephropathy, amyloid, HIVAN, and MN with RVT. MN may be the presenting finding in cases of SLE, whereas diffuse proliferative GN with low C3, when associated with SLE, occurs in patients who also have signs and symptoms of joint, skin, or systemic organ involvement. A cause of very early MN is an idiosyncratic reaction to NSAIDs, even when taken in small amounts. Many other drugs including penicillamine, mercury, and gold can cause MN. Proteinuia from any cause may present earlier, or increase, when DM, obesity, partial nephrectomy, or increased renal venous pressure exists.

Solid tumors, particularly lung, gastric, intestinal, bladder, breast, and prostate, may underlie a paraneoplastic syndrome of MN. While it is important to take a history and examine the patient carefully for signs of cancer, it is not the rule to work up every patient with MN exhaustively for cancer. But the neoplastic nephropathy may be a sign of recurrent disease and may even be recognized by tumor antigens. MN also occurs in the context of many infectious diseases, including syphilis, malaria, schistosomiasis, and hepatitis B. Recently, 70% of patients with primary MN have been found to have circulating levels of an autoantibody to the phospholipase A2 receptor (anti-PLA2R) in the subepithelial region of the glomerular basement membrane.

Patients who have a nephrotic and nephritic sediment in childhood may have membranoproliferative glomerular nephritis (MPGN) with low C3 as a primary disease (C3 glomerulopathy or dense-deposit disease). The adult presenting with a nephrotic and nephritic picture should be considered to have an immune complex disorder unless proven otherwise. In amyloidosis there may be a simultaneous and severe tubular syndrome in addition to the albuminuria, consisting of nephrogenic diabetes insipidus and a hyperkalemic RTA. When NS is associated with Fanconi's syndrome (glycosuria, phosphaturia, uricosuria, and aminoaciduria), one must consider multiple myeloma in the adult.

HEMATURIA AND LOWER URINARY TRACT SYNDROMES

Because blood could enter the urinary tract from any structure, identifying the source is a necessary first step. As a rule, hematuria associated with flank pain or ureteral colic is more characteristic of a lower urinary tract source, such as a stone or an obstructing lesion. However, when flank pain without colic is noted, one has to consider swelling of the kidney and stretching of the renal capsule, as might be seen in acute urinary obstruction, kidney infection, interstitial nephritis, and, on occasion, acute glomerular nephritis. Heavy bleeding with clots is more characteristic of lower tract bleeding but could result from trauma to the kidney. In patients who have acute flank pain and hypertension, assuming the pain is due to a kidney stone may miss an occlusive vascular event from an embolus or in situ clot within the arterial blood supply. Patients who engage in strenuous exercise may develop hematuria that can be ruled out by retesting the urine after a week or two of rest. True hematuria can be distinguished from hemoglobinuria and myoglobinuria by urinalysis.

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TABLE 308-2 Hereditary and Congenital Diseases

HEREDITARY GLOMERULAR DISEASES: NEPHROTIC SYNDROME OR RENAL FAILURE	HEREDITARY TUBULAR DISEASES	APOL1	CILIOPATHIES	CHANNELOPATHIES	CONGENITAL ANOMALIES
Podocyte mutations; basement membrane mutations; hereditary nephritis; X-linked and somatic mutations; *COL4A5*; Allport's syndrome with hematuria neurosensory hearing loss, conus lentus; Fabry's disease (X-linked alpha-galactosidase deficiency)	Fanconi's syndrome with renal glycosuria, proximal tubule RTA, hypophosphatemia, hypouricemia; distal RTA with hearing loss, renal calcification; nephrogenic diabetes insipidus; hyper- or hypotensive disorders with hypo- or hyperkalemia; metabolic alkalosis or acidosis	Mutant allele in African-American population, leading to susceptibility to further injury in many other disorders including diabetes mellitus	Autosomal dominant and recessive polycystic kidney disease; medullary cystic kidneys (metabolic acidosis and salt-wasting); other phenotypes	Gitelman's and Bartter's syndromes; disorders of eNaC and proton pumps	Unilateral agenesis; dysgenetic disease; medullary sponge kidney (hematuria, stones, or infection); horseshoe kidneys (proteinuria); malposition; ureteropelvic obstruction; cystoureteral reflux (proteinuria)

Symptoms of lower urinary tract disorders are dysuria, urinary frequency, urgency, incomplete emptying, hematuria that is most pronounced at the beginning of the stream, as well as a poor urinary stream. Among these disorders are urinary tract infection and prostatic hypertrophy. Hematuria of recent onset noted in the fourth decade of life or beyond should undergo a urologic evaluation and most likely cystoscopic examination of the bladder. However, chronic hematuric conditions may be associated with chronic glomerular disease such as thin basement membrane, hereditary nephritis, or IgA nephropathy. That is why it is so important to search for previous examinations such as insurance, military, or employment exams done in the past to determine if hematuria was present in order to establish chronicity.

ACUTE KIDNEY INJURY

Acute kidney injury (AKI) is defined by the retention of nitrogenous solutes such as urea, uric acid, and creatinine. The creatinine concentration goes up primarily related to the amount of water absorbed along the nephron. Because the urea but not creatinine has a reabsorptive component, the clearance of urea is less than the clearance of creatinine. With a constant catabolic rate, a decrease in glomerular filtration alone will affect the BUN and creatinine proportionately. In contrast, a disorder associated with enhanced urinary concentration (high ADH), in which urea as well as water has increased reabsorption in the process of concentrating the urine, the retention of urea nitrogen exceeds that of creatinine, accounting for an elevated BUN-to-creatinine ratio. Oliguric or oligoanuric states are often characterized as *prerenal*, meaning preglomerular vascular disease or low perfusion states; *renal*, indicating intrinsic renal disease; or *postrenal*, indicating obstructive nephropathy or uropathy (see Table 308-1).

Prerenal states are often related to a decreased perfusion pressure to the glomerular capillaries or some other interference with filtration. The differential diagnosis includes proliferative GN (the bloodless capillary); a preglomerular vascular disease such as scleroderma or thrombotic microangiopathy; the use of pressors for the maintenance of blood pressure; vasoconstrictive substances including cocaine, iodinated IV contrast media, hemoglobin or myoglobin, and certain antibiotics like vancomycin, cyclosporine, and tacrolimus (calcineurin inhibitors); other vasoconstrictive drugs like nonsteroidals; and renal nerve stimulation. Hypercalcemia and hypoxia are also vasoconstrictive. Correcting hypoxemia has a diuretic effect.

Intrinsic Renal Disease All of these will decrease perfusion pressure and decrease GFR while stimulating sodium and water reabsorption. Therefore, the characteristic urinary findings are low urinary sodium excretion and concentrated urine. The urine sodium concentration is usually low and the urine to plasma creatinine concentration is usually >40. The same result could occur in bilateral renal artery stenosis, heart failure, and other circulatory collapse. If damage is done to the renal tubules, so-called **intrinsic renal disease**, there may or may not be oliguria and in fact there could be polyuria (unresponsiveness to ADH), sodium-wasting, potassium-wasting, and acid-base disorders, each of which may predominate. One of the most frequent causes of oliguric acute renal failure is acute tubular necrosis (ATN), caused by gram-negative sepsis, hemodynamic collapse, hemoglobinuria or myoglobinuria, and medication or toxin ingestion. Examples are crush injury, heroin and alcohol stupor, and compartment syndrome. Drug causes include NSAIDs, aminoglycosides, and chemotherapeutic drugs like cisplatinum and methotrexate.

Postrenal States Postrenal states accompanied by anuria suggest complete obstruction to the flow of urine, while polyuria and bland urinary sediment suggest incomplete obstruction combined with ADH unresponsiveness. The fractional excretion of sodium is usually >1–2%, whereas in oliguric prerenal syndromes the fractional excretion of sodium is usually <1%. Exception to the low sodium excretion would be in the case of hyperchloremic metabolic alkalosis, where chloride is being conserved and sodium excreted with bicarbonate.

Urinary tract obstruction is usually characterized by hydronephrosis visualized on ultrasound images or large volumes of urine removed by bladder catheterization, indicating bladder outlet obstruction. Exceptions to finding hydronephrosis on an imaging study include the extrinsic obstructive compression of the kidney or ureter in retroperitoneal disease; cases in which imaging was done too soon, for example <4 days after obstruction; or when the patient was simultaneously obstructed and volume depleted.

As mentioned earlier, acute tubular diseases that are not primarily due to inflammation can be caused by intratubular obstruction, from hyperoxaluric states, or by uric acid or phosphates in tumor lysis syndrome following chemotherapy for lymphoproliferative disorders. Other drug therapies that cause tubular injury include cisplatinum, ifosfamide, methotrexate, aminoglycosides, and amphotericin B, and toxicity through exposure to heavy metals.

Kidney injury in the setting of systemic disease with dysfunction in other organs includes the cardiorenal, pulmonary renal, and hepatorenal syndromes. Other syndromes that typically involve the kidney include febrile illness, systemic infection with septic shock, hemodynamic collapse due to cytokines, or capillary leak syndrome. These syndromes as well as the endothelial origins of thrombotic microangiopathies and hemolytic uremic syndrome are shown in Fig. 308-4. Obstetric patients with abruptio placenta may develop bilateral cortical necrosis and as a consequence, anuria, and often irreversible renal failure. In addition, patients with atheroembolic syndrome can present with AKI or chronic progressive damage following an aortic catheterization. If the blood vessels catheterized involve the aortic arch, an encephalopathic syndrome that can mimic uremic encephalopathy may follow. As many as 70% of patients with COVID-19 who required ventilators during the early phase of the pandemic developed multiorgan failure, including AKI. AKI may also be a risk for chronic kidney disease following a COVID-19 infection.

CHRONIC KIDNEY DISEASE AND THE UREMIC SYNDROME

Chronic kidney disease (CKD), progressing over months to years, can be the long-term result of any of the previously mentioned diseases. If the time course is unknown, other ways to diagnose CKD versus

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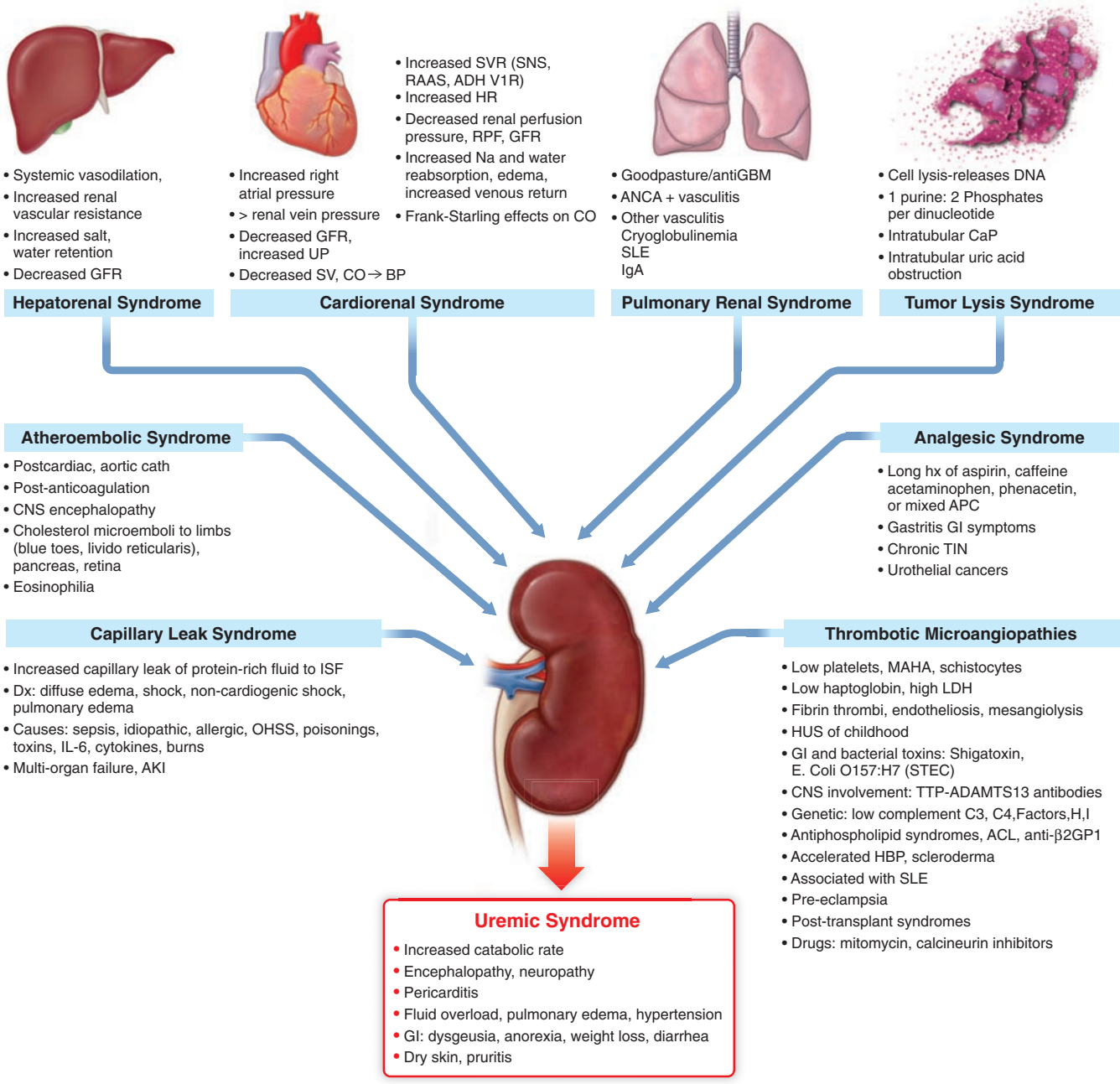


FIGURE 308-4 Categories of systemic or non-renal-based syndromes that involve the kidney and in advanced cases can cause the uremic syndrome requiring renal replacement therapy. ACL, anticardiolipin antibodies; ADH V1R, antidiuretic hormone V1 receptor; anti-β2GP1, anti-beta-2-glycoprotein-1; APC, aspirin phenacetin caffeine; BP, blood pressure; CaP, calcium phosphate crystals; CO, cardiac output; HUS, hemolytic uremic syndrome; IgA, immunoglobulin A; ISF, interstitial fluid; LDH, lactic dehydrogenase; MAHA, microangiopathic hemolytic anemia; OHSS, ovarian hyperstimulation syndrome; RAAS, renin angiotensin aldosterone system; RPF, renal plasma flow; SV, stroke volume; SNS, sympathetic nervous system; SVR systemic vascular resistance; TIN, tubulointerstitial nephritis; TTP, thrombotic thrombocytopenic purpura; UP, urine protein.

a more recent kidney disorder are useful, for example, the finding of bilaterally small kidneys on a renal ultrasound. Kidneys by this method are normally 10–12 cm in length. Small kidneys, <8 cm, are likely atrophic with irreversibly low function; but in some cases of CKD, such as diabetes, the kidneys may be large despite kidney failure. Remarkably, even atrophic kidneys may still produce renin to sustain blood pressure and erythropoietin to minimize anemia; however, due to low calcitriol in CKD, patients develop secondary hyperparathyroidism. The normal parathyroid gland weighing 25 mg would be limited in raising PTH, such that very elevated PTH likely is in favor of CKD. Thinning of the renal cortex is also a sign of chronicity.

DM accounts for approximately 50% of CKD patients that progress to end-stage renal disease (ESRD). Other important causes leading to ESRD are IgA nephropathy, ischemic nephropathy without proteinuria manifesting after age 50 years with hypertension, and signs of other large-vessel disease, such as intermittent claudication from peripheral vascular disease, stroke, and coronary artery disease. In contrast, renal arterial disease due to fibromuscular dysplasia, seen predominantly in white women in the fourth decade of life, does not progress to renal failure and is a treatable form of hypertension. Other arterial diseases include polyarteritis nodosum (PAN) and, in medium-sized vessels, a Kawasaki-like illness affecting children, which has had a thirtyfold increase since the onset of COVID-19 as part of the multisystem inflammatory syndrome.

Hereditary and congenital diseases of the kidney are shown in Table 308-2. Autosomal dominant PCKD occurs in all ethnic populations. The patient may first show signs of cysts in late teenage years with hypertension, urinary or cyst infection, or pain and bleeding. Kidney stones may be seen. By the age of 30 years, individuals with the disorder will have ultrasound-detectable cysts. The clinician should know that

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TABLE 308-3 Stages of Chronic Kidney Disease

STAGES OF CKD, ABNORMALITY > 3 MONTHS	RANGE OF eGFR	CLINICAL FEATURES	IMPORTANCE
1	>60 mL/min/1.73 m^2	Abnormal urinalysis, abnormal renal imaging	Risk of progression to later stages increases with increased proteinuria and dependent on cause of disease
2	>60 mL/min/1.73 m^2	Abnormal urinalysis, abnormal renal imaging	Mild risk of progression to later stages increases with increased proteinuria and dependent on cause of disease
3a	45–60 mL/min/1.73 m^2	Cardiovascular disease or other organ damage	Moderate risk of progression of disease. Pay attention to other vascular risk factors, high BP, lipids, smoking, weight
3b	30–45 mL/min/1.73 m^2	Proteinuria	High risk of progression
4	15–30 mL/min/1.73 m^2		High likelihood of progression to ESRD, need preparation and education regarding choices for RRT including transplant and dialysis
5	<15 mL/min/1.73 m^2		Highest risk of requiring RRT

a diagnosis of PCKD requires multiple cysts in both kidneys. Cysts are also found in the liver. The first manifestation of PCKD may be a cerebral hemorrhage, since berry aneurysms are present in a small subset of patients with this disease, most commonly in the circle of Willis.

Hereditary nephritis with many variations includes the Alport syndrome, involving a gene mutation (*COL4A5*, accounting for the majority of cases) effecting the alpha 3 region of the noncollagenous domain of type IV collagen in GBM. There is an X-linked pattern of genetic transmission, where males may be most severely affected by renal failure in middle age. The syndrome involves hematuria, neurosensory hearing loss, and ocular deformities of the lens. A predominant symptom in hereditary nephritis is microscopic hematuria and the pathologic correlate to this is thin basement membranes. It must be distinguished from the less severe and more common related condition known as thin basement membrane disease as a cause of hematuria.

Certain congenital abnormalities of the kidney, such as horseshoe and ectopic kidneys, also shown in Table 308-2, consist of low nephron numbers that may result in secondary FSGS. The condition known as ureteral reflux is an abnormality involving the functional insertion of the ureter into the muscular bladder, which normally closes off the ureter upon bladder contraction. In reflux disease the ureter cannot prevent retrograde flux of urine during micturition. In children this may cause hypertension and, in some cases, secondary FSGS. Surgical correction of the abnormality may be curative, but the child may outgrow it in adulthood in any case.

The stages of CKD are listed in **Table 308-3** and are important in terms of planning for renal replacement therapy (dialysis and transplantation), and for assessing and slowing the rate of progression, defined as the decline in GFR over time. In CKD, the estimated GFR (eGFR) is determined by the current stable serum creatinine, entered into one of several equations derived from clinical trial data that compared the subject's value to actual measurements of GFR. For example, the Modification of Diet in Renal Disease (MDRD) study used a radioisotope (I^{125}iothalamate) clearance to measure true GFR. The equation contains additional factors such as age and sex to derive the current eGFR, corrected for 1.73 m^2 body-surface area. This value of eGFR is *not* a creatinine clearance, nor a measurement of GFR.

There has been debate about the inclusion of corrective factors that would alter the result from that equation, based on the patient's ethnicity and race. For example, it has been recommended in some centers to multiply the eGFR value by a factor of 1.2 in black patients in order to correct for the perceived underestimate of GFR using the MDRD equation. There is considerable objection to making this correction, which may have the effect of introducing bias based on race. Because the corrected eGFR will give a higher value to the GFR, there could be delays in discussion of renal replacement therapy and in initiation of treatment. Consequently, the correction has fallen out of favor. Recently an alternative, the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI), has been used that excludes race as a parameter. Another method of calculating GFR uses cystatin-C.

In the course of CKD, the clinician must evaluate the patient for symptoms and signs of worsening renal function. When the eGFR is <15–20 mL/min, the patient should already have had conversations about choices of eventual treatment and, if indicated, surgical preparation for peritoneal or hemodialysis. Attention should be given to the sensitive topic of transition to a lifesaving but mechanical procedure. It is also important to discuss transplantation and issues of timing. The ethical standard is for separate clinicians to evaluate the recipient and donor of a kidney transplant. Confidentiality and privacy are essential. Very importantly, access to transplantation has not been equally offered to minority populations and it is incumbent on the clinician to avoid bias by discussing the option with every patient.

Symptoms of CKD are often nonspecific and include fatigue, weakness, loss of appetite and taste, weight loss, mood changes, and metabolic encephalopathy that may involve cognitive changes such as decline in executive function and the ability to calculate and remember. Other symptoms are peripheral and autonomic neuropathies and sleep and movement disorders such as restless legs and asterixis or myoclonus. Patients with AKD lose their ability to concentrate the urine, and therefore have an obligate urine output of approximately 2 L. This results in nocturia, since the bladder fills during the night. Ironically, the loss of nocturia may be sensed by the patient as an improvement of symptoms but actually indicates worsening oliguria. Pruritis commonly occurs. Clinical issues include changes in medications and doses. The full uremic syndrome is characterized by severe hypertension, the above-mentioned encephalopathic manifestations, GI bleeding, pericarditis, severe electrolyte disturbances, particularly hyperkalemia, and secondary hyperparathyroidism. Anemia is frequently present and though it may be due to iron deficiency or decreased erythropoietin, other causes of anemia should be investigated. Patients respond well to erythropoietic stimulating agents, with the goal of therapy >10 g/dL but <12 g/dL. Notably, not all patients are anemic. For example, some patients with urinary tract obstruction, polycystic kidney disease, renal vascular disease, and renal cell carcinoma may in fact have erythrocytosis.

In nephrotic syndrome, heavy proteinuria may persist throughout the course of CKD and carries a worse prognosis. In all kidney diseases, including the nephritic processes, renal disease may progress to late stages even when the acute inflammation has subsided. However, the underlying mechanisms of progressive renal failure involve inflammatory pathways (reflected in elevated levels of C-reactive protein and erythrocyte sedimentation rate in CKD), emphasizing the importance of managing risk factors for cardiovascular disease, such as hypertension, smoking, hyperlipidemias, diabetes, and obesity early in the course of CKD.

Some chronic renal syndromes may present with predominant tubular dysfunction, including genetic diseases known as channelopathies. Examples are the depletion of electrolytes such as phosphorus and potassium and proximal RTA due to Fanconi's syndrome, mentioned above; hypokalemic alkalosis and volume depletion in Bartter's syndrome (mutations resulting in abnormal loop of Henle function); and electrolyte losses in Gitelman's, a defect in the thiazide-sensitive sodium-chloride transporter. Gitelman's also causes hypochloremic hypokalemic metabolic alkalosis. Bartter's and Gitelman's can be distinguished by examining the urine for the usual solutes regulated by these transporters. Thus, Bartter's is associated with hypercalciuria

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and the inability to put out a concentrated urine, whereas Gitelman's is associated with hypocalciuria and preserved ability to concentrate the urine. Fanconi's syndrome results in hypokalemic acidosis and can be distinguished from distal RTA, which does not associate with the loss of glucose, amino acids, and phosphate in the urine.

Disorders involving the collecting duct and its transporters of sodium, potassium, and acid base can result in syndromes that also suggest certain diagnoses. For example, polyuria due to inappropriate free water losses and insensitive to ADH is characteristic of nephrogenic diabetes insipidus and it is common to see simultaneous defects in potassium secretion and hydrogen ion secretion, with hypertension suggesting blockade of one of the sodium regulators of ENaC. These disorders are usually associated with hyporenin and/or hypoaldosterone states, which include UTO, type 4 RTAs in DM, amyloidosis, or Addison's disease. In contrast, hypokalemic metabolic alkalosis with volume expansion may suggest an adrenal adenoma, unilateral renal artery stenosis, ACTH-secreting tumors, licorice abuse, or potassium-sparing diuretics.

Renal Masses Patients with renal cell carcinoma may have been diagnosed via an incidental finding in an abdominal imaging study, or sometimes by a palpable mass best felt with the patient supine, usually a thin patient but occasionally a very large protruding mass. Very large masses or multiple masses that are easily palpable may represent cystic diseases of the kidney, including polycystic kidney disease (PKD) or even a single cyst, versus a congenital ureteral pelvic obstruction. Other times, a renal cell carcinoma can present as anemia, possibly caused by hematuria, or as back pain associated with metastatic lytic vertebral lesions. Metastases may involve the lungs and the bone marrow as well.

Imaging and Renal Biopsy Indications For hematuric syndromes, imaging may add valuable information, particularly in the patient who has heavy bleeding or blood clots in the urine. Renal pathology may be detected as an abdominal mass, as in the case of renal cell carcinoma, chronic urinary tract obstruction, or cystic diseases of the kidney including polycystic kidney disease (PKD) and simple cyst. If the patient has a known history of tubular sclerosis or the finding of skin fibroadenoma, one might identify a renal mass found on CT imaging as an angiomyolipoma.

The renal ultrasound is efficacious in determining the size and symmetry of the kidneys and in excluding urinary obstruction. It is helpful in detecting renal cysts or mass, but less useful in kidney stone disease. Ultrasound is not as accurate a tool for angiomyolipomas. The renal-limited noncontrast CT scan is the standard test for nephrolithiasis but carries the risk of accumulative radiation. MRI is often useful in evaluating and following renal masses, including renal cell carcinoma. The patient with renal disease may develop the complication of systemic sclerosis after receiving gadolinium; new contrast media to replace it are emerging. CT iodinated contrast media remain a problem, particularly in the patient with vascular disease of the kidney. Radioisotope scanning is useful in demonstrating the percentage of renal function coming from each kidney. Finally, in many of the diseases discussed above, diagnosis ultimately depends on renal biopsy and pathologic evaluation.

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309 Cell Biology and Physiology of the Kidney

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The kidney is one of the most highly differentiated organs in the body. At the conclusion of embryologic development, nearly 30 different cell types form a multitude of filtering capillaries and segmented nephrons enveloped by a dynamic interstitium. This cellular diversity modulates a variety of complex physiologic processes. Endocrine functions, the regulation of blood pressure and intraglomerular hemodynamics, solute and water transport, acid-base balance, and removal of drug metabolites are all accomplished by intricate mechanisms of renal response. This breadth of physiology hinges on the clever ingenuity of nephron architecture that evolved as complex organisms came out of water to live on land.

EMBRYOLOGIC DEVELOPMENT

Kidneys develop from intermediate mesoderm under the timed or sequential control of a growing number of genes, described in Fig. 309-1. The transcription of these genes is guided by morphogenic cues that invite two ureteric buds to each penetrate bilateral metanephric blastema, where they induce primary mesenchymal cells to form early nephrons. The two ureteric buds emerge from posterior nephric ducts and mature into separate collecting systems that eventually form a renal pelvis and ureter. Induced mesenchyme undergoes mesenchymal epithelial transitions to form comma-shaped bodies at the proximal end of each ureteric bud leading to the formation of S-shaped nephrons that cleft and enjoin with penetrating endothelial cells derived from sprouting angioblasts. Under the influence of vascular endothelial growth factor A (VEGF-A), these penetrating cells form capillaries with surrounding mesangial cells that differentiate into a glomerular filter for plasma water and solute. The ureteric buds branch, and each branch produces a new set of nephrons. The number of branching events ultimately determines the total number of nephrons in each kidney. There are ~900,000 glomeruli in each kidney in normal-birth-weight adults and as few as 225,000 in low-birth-weight adults, with the latter producing numerous comorbid risks.

Glomeruli evolve as complex capillary filters with fenestrated endothelia under the guiding influence of VEGF-A and angiopoietin-1 secreted by adjacently developing podocytes. Epithelial podocytes facing the urinary space envelop the exterior basement membrane supporting these emerging endothelial capillaries. Podocytes are partially polarized and periodically slough into the urinary space by epithelial-mesenchymal transition and, to a lesser extent, apoptosis, only to be replenished by migrating parietal epithelia from Bowman capsule. Impaired replenishment results in heavy proteinuria. Podocytes attach to the basement membrane by special foot processes and share a slit-pore membrane with their neighbor. The slit-pore membrane forms a filter for plasma water and solute by the synthetic interaction of nephrin, annexin-4, CD2AP, FAT, ZO-1, P-cadherin, podocin, TRPC6, PLCE1, and Neph 1-3 proteins. Mutations in many of these proteins also result in heavy proteinuria. The glomerular capillaries are embedded in a mesangial matrix shrouded by parietal and proximal tubular epithelia forming Bowman capsule. Mesangial cells have an embryonic lineage consistent with arteriolar or juxtaglomerular cells and contain contractile actin-myosin fibers. These mesangial cells make contact with glomerular capillary loops, and their local matrix holds them in condensed arrangement.

Between nephrons lies the renal interstitium. This region forms a functional space surrounding glomeruli and their downstream tubules, which are home to resident and trafficking cells such as fibroblasts, dendritic cells, occasional lymphocytes, and lipid-laden macrophages. The cortical and medullary peritubular capillaries, which siphon off